

This technology is based on an existing piece of technology called streak camera, which is an ultra-fast device for measuring intensity variation of pulse of light in a given time. While streak cameras are able to record in one dimension only, new algorithms and components can record in two dimensions. After acquiring raw data, actual images are formed on a computer, using a technology called computational imaging. The camera's immediate application is in biomedicine, and it can be used to detect diseases and reflect cellular environmental conditions such as pH or oxygen pressure.

A machine that can sniff better than dogs

A team of scientists at Duke University has developed a new method for sniffing toxic gases. This technology, which is more accurate than sniffer dogs, is based on terahertz radiations (T-rays). Scientists claim that with this technology, the presence of nerve gas within one kilometre can be detected.

Terahertz radiations are a form of invisible light, which is more sensitive than sensors even below normal atmospheric pressure. These rays are already being used to detect tumours and for screening at airports.

The research team is led by Henry Everett from Duke University. He believes that, this technology can have a high impact on security at military areas. It will become very easy to



This machine can be set up around the perimeter of an area where soldiers are living, as a kind of trip wire for nerve gas

detect nerve gas in a specific perimeter as T-rays can be zapped with gas molecules in order to create a unique barcode-like fingerprint spectrum.

The team performed a test in which they blasted a cloud of gas with two-terahertz beam

of similar energy as gas molecules and an infra-red laser. The experiment was similar to whacking a molecule with an infra-red sledge hammer. Scientists had to tune the wavelength of each beam to match with the type of molecule, and any change indicated the presence of gas.

The team is currently working on the detection of toxic industrial chemicals like ammonia, carbon disulfide, nitric acid and sulphuric acid. This technology can be used by law enforcement officers to detect toxic gases coming from meth labs and other drug-related illegal activities.

Emotion-sensing technology that can detect mood

Dell's emotion-sensing technology, the pet project of Jai Menon, chief research officer and IBM's former chief technology officer, attracted a lot of attention at TED talk in Amsterdam.

According to Menon, his team has been working to gain more knowledge about the intention of an individual when he or she is using his laptop or mobile phone.

The software can detect when a gamer is feeling bored with the game. If he or she is, the challenge level of the game will be adjusted accordingly and related feedback will be given to the gaming company. It is all about personalisation through user's intent.

Dell's emotion-sensing technology is similar to GPS. Dell is looking at camera based inputs to identify your state of mind, and stress level will be detected through a speech-analysis software. So, it is all about your mood that is in focus right now.

The technology can also be beneficial to healthcare sector. For example, Menon added, before a heart-attack, a person shows certain indications. This technology will be able to detect those indications and their aftermath to control the worst situation.

Bluetooth 4.2: the foundation of the IoT

Bluetooth 4.2 and its specifications have been revealed that clearly indicate that it is meant for the world where everything is connected by the Internet. However, Bluetooth is not just about the Internet. It is time for Bluetooth 4.2 to step into action to make the Internet of Things (ToT) work.

Bluetooth Special Interest Group, which develops new standards for wireless technology, has officially adopted version 4.2 of Bluetooth core specification. The new Bluetooth can increase privacy, speed and bring IP connectivity to all devices that support it. It means sensors and other smart devices will connect to the Internet directly via Bluetooth. Home-automation will be easier as smart products like light bulbs, thermostats and door sensors will be able to access the Internet without any mediating device. Hence, Bluetooth 4.2 is the foundation for the Internet of Things.

This technology is said to be 2.5 times faster than its predecessor and will provide developers with the flexibility to innovate. It is estimated that by 2020, around 28 billion devices will be connected to the Internet, including cars, toasters, guitars and shavers, among others. The new Bluetooth 4.2 technology will connect these devices to the Internet using Bluetooth via a router or an access point without any intermediate device.

Privacy and security issues are at the top of the technology. It addresses the loopholes in its previous specifications and makes it almost impossible to track any device via Bluetooth without user permission. Power consumption is also low, which means, Bluetooth will no more drain your gadget's battery.



Dell laptops can detect your mood using emotion-sensing technology